

Elucidating Surface and Bulk Phase Transformation in Fischer–Tropsch Synthesis Catalysts and Their Influences on Catalytic Performance

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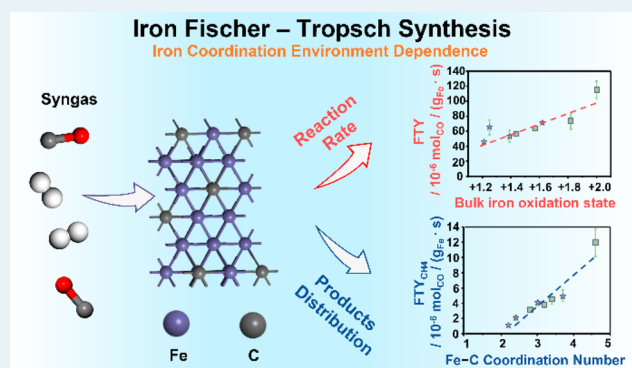
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Supporting Information

ABSTRACT: Understanding the structure–performance relationship for Fe-based Fischer–Tropsch synthesis (FTS) catalysts is crucial for the design of improved FTS catalysts. Herein, catalysts derived from four pure phases of iron (i.e., nanosized Fe_2O_3 , $\chi\text{-Fe}_5\text{C}_2$, $\theta\text{-Fe}_3\text{C}$, and Fe^0) were obtained under controlled conditions. Combining various surface and bulk characterization techniques, we revealed that core–shell structures were formed for all catalysts irrespective of the catalyst precursors, while the compositions within the core and shell strongly depend on the intrinsic properties of the precursors employed. Most importantly, we report that the iron coordination environment mainly governs the performance during FTS. The iron time yield is found to correlate linearly with the average bulk iron oxidation state rather than the surface properties, and the CH_4 yield is related to the coordination number of the Fe–C bond. This study thus provides general descriptors to interpret the structure–performance relationships of Fe-based FTS catalysts.

KEYWORDS: Fischer–Tropsch synthesis, iron catalysts, structure–performance relationship, descriptor, iron oxidation state, Fe–C coordination number



INTRODUCTION

The Fischer–Tropsch synthesis (FTS) is a process used to convert syngas directly to hydrocarbon fuels and chemical feedstocks,¹ which offers feedstock flexibility to meet a rapid increase of energy demand.² Currently utilized, iron-based catalysts exhibit significant advantages including their low price, chemical robustness, high tolerance for the wide H_2/CO ratios obtained for H_2 -deficient syngas obtained from coal, and good biomass conversion. Generally, iron oxides have been used as catalyst precursors for FTS, undergoing reduction and carburization during activation and reaction. But a complex mixture of phases (e.g., Fe oxides and carbides) is often experimentally identified under FTS conditions. Researchers including de Smit et al. claimed that the stability and interconversion of these carbides strongly depend on the carbon chemical potential and reaction temperature.³ Furthermore, the carbide phases could be reoxidized in a syngas atmosphere. The rich complexity of iron species and the dynamic nature of the iron–carbon–oxygen system under the FTS environment makes it challenging to identify and deconvolute the various factors affecting the structure–performance relationship in these systems.^{4–6} The air-

sensitivity of iron species also puts forward higher requirements for characterization techniques.⁴ Despite tremendous effort being made to elucidate the role of different iron carbides, the results presented in the literature are somewhat controversial. The Hägg carbide ($\chi\text{-Fe}_5\text{C}_2$) is mostly commonly regarded as the primary active species, as evidenced by the relationship between $\chi\text{-Fe}_5\text{C}_2$ and catalytic activity in a number of cases.^{3,6,7} Fe_2C is predicted to be more active in CO dissociation and hydrogenation by density functional theory (DFT) calculations, but it is rarely studied due to its thermodynamic instability.^{8–10} $\theta\text{-Fe}_3\text{C}$ has traditionally been considered as a deactivation phase;^{3,5} however, Liu et al. have found it contributes to the high selectivity of lower olefins.¹¹ More recently, Fe_7C_3 was found to have the highest turnover frequency compared to the $\varepsilon\text{-Fe}_2\text{C}$ and $\chi\text{-Fe}_5\text{C}_2$ phases.¹² Therefore, the speciation of iron carbides goes beyond the single-site paradigm, which includes chemical transformation and is influenced by precursor composition, synthesis method,

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Scheme 1. Illustration of Sample Collection under Two Conditions

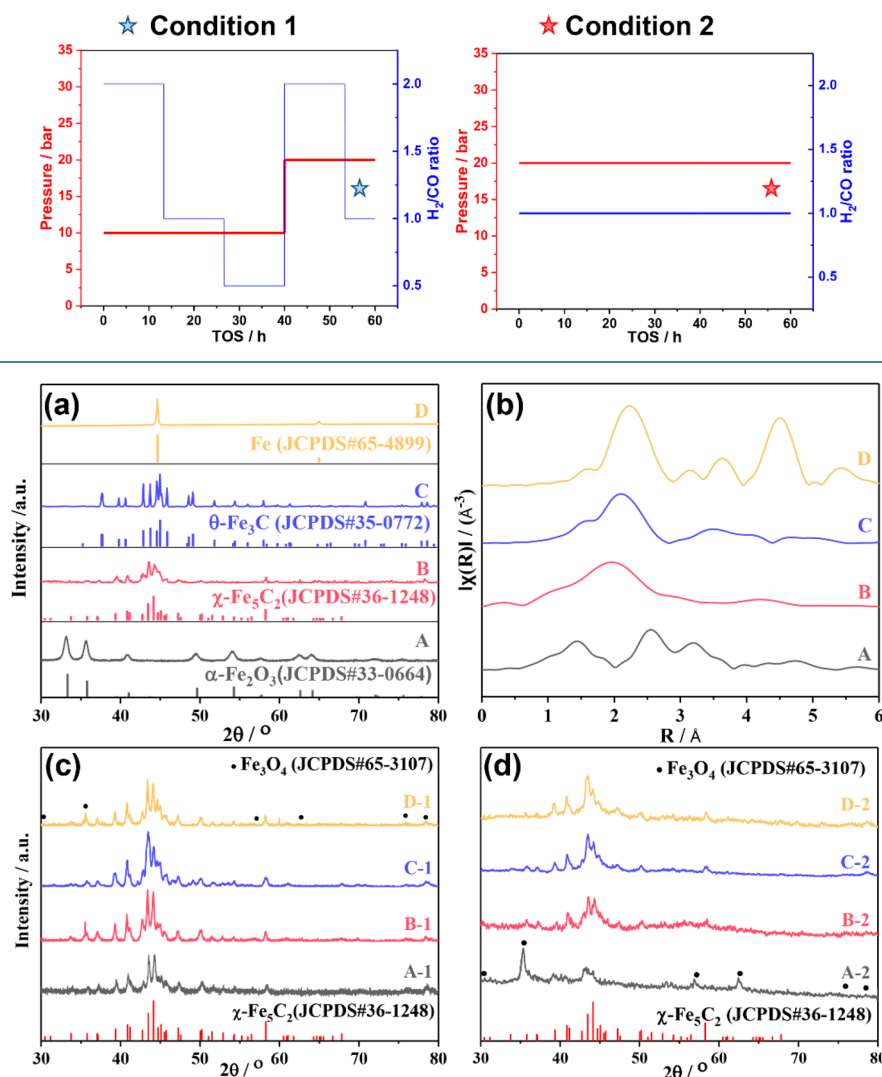


Figure 1. Characterization of catalysts precursors and used catalysts: (a,b) experimentally obtained XRD patterns and k^2 -weighted Fourier-transformed Fe K-edge EXAFS data of precursors; (c,d) XRD patterns of used catalysts under condition 1 and condition 2.

pretreatment, and reaction conditions. Besides the complexity and instability of iron phase, the structure-sensitivity of FTS reaction also plays an important role in catalytic performance. Size-dependence has been extensively reported by experimental work, which affects the coordination environment of Fe atoms,^{13,14} and different planes of a certain iron phase also show different CO activation and C–C coupling abilities in the DFT study.^{15–17} Third, the heterogeneous structure of Fe catalysts from surface to bulk is observed under FTS atmosphere.^{18–21} These issues make it challenging to interpret the structure–activity relationship generally. On the other hand, the products selectivity in FTS is of great scientific and industrial importance. Generally, the products distribution is a compromise between C–C coupling and secondary hydrogenation, which is closely related to the chemical properties of the Fe species. Insight into the relationship between the Fe species and selectivity is highly desirable for designing selective catalysts of FTS.

Herein, we initiated a study with four pure phases of iron (Fe₂O₃, χ -Fe₅C₂, θ -Fe₃C, and Fe⁰; denoted as A, B, C, and D, respectively) as precursors, and collected two series of derived

catalysts at 60 h under the FTS conditions ($T = 340^\circ\text{C}$, $\text{H}_2/\text{CO} = 1$, 20 bar, $\text{WHSV} = 30\,000\text{ mL}/(\text{g}_{\text{cat}}\cdot\text{h})$) with the Flowrence high-throughput unit.²² As shown in Scheme 1, series 1 catalysts (denoted as A-1, B-1, C-1, and D-1) were subject to lower pressures and varied H_2/CO ratios before reaching the FTS conditions; series 2 catalysts (denoted as A-2, B-2, C-2, and D-2) were pretreated by keeping the FTS conditions. By means of comprehensive and complementary “quasi in situ” characterization, we aimed at correlating the surface and bulk properties of the intermediate samples with both activity and selectivity.

RESULTS AND DISCUSSION

As shown in Figure 1a,b, we successfully synthesized isolated nanosized iron precursors with a high degree of crystallinity, as evidenced by X-ray diffraction (XRD) and extended X-ray absorption fine structure spectroscopy (EXAFS). Under condition 1 (Figure 1c), all the samples obtained show the characteristic diffraction pattern of χ -Fe₅C₂ (c.f. JCPDS no. 36-1248), indicating χ -Fe₅C₂ is the dominant iron phase. But when under condition 2 (Figure 1d), the A-2 catalyst showed

catalyst, the Fe_3O_4 phase mainly lies in the shell while the iron carbides are predominant in the core. The boundary between the core and shell is more ambiguous for the A-1 than that of the C-1 catalyst. The different density of carbides compared with oxides (5.2 g/cm^3 for Fe_2O_3 and Fe_3O_4 vs 7.57 g/cm^3 for Fe_5C_2) might lead to the interlayer shrink for A-1 catalyst, while interlayer expansion for the C-1 catalyst. In other samples the TEM images under two conditions that show similar core-shell patterns can be found in Figure S1, and the specific compositions in each part of the particles are strongly dependent on the starting precursors. The peaks at ~ 286.0 and $\sim 296.0 \text{ eV}$ shown in the EELS profile (Figure 2) are ascribed to the C $\text{K}-\pi$ ($1s$ to π^* transition, which indicates the presence of sp^2 hybridized carbide species) and the C $\text{K}-\sigma$ ($1s$ to σ^* absorption peak, which stands for amorphous carbon species).^{19,20} For A-1 (Figure 2a,e), spot 1 shows the carbon $\text{K}-\sigma$ signal, the O K -edge signal as well as the Fe L-edge signal. Combining this with the lattice spacing from HRTEM, it implies that the iron oxide coexists with amorphous carbon species. For spots 2 and 3, C $\text{K}-\pi$, $\text{K}-\sigma$ signals, O K -edge, and Fe L-edge signals are observed, indicating the coexistence of iron carbide(s) and Fe_3O_4 phases in these regions. Note that the relative $\text{K}-\pi/\text{K}-\sigma$ intensity ratio ($I_{\text{K}-\pi}/I_{\text{K}-\sigma}$) increases along the spectrum scan line from spot 2 to spot 3, indicating the relative iron carbide composition increases along the core to the shell. For spot 4, only C $\text{K}-\sigma$ signals were observed, revealing the Fe particles are entangled by amorphous carbon deposits. For sample C-1 (Figure 2f,j), the main differences lie in the core region in Figure 2j from the iron carbide where the $I_{\text{K}-\pi}/I_{\text{K}-\sigma}$ decreases from the core to the shell (spot 5 to 6 and 7). The existence of amorphous FeC_x species can be confirmed by the fingerprint-like whorls across the whole region. X-ray photoelectron spectroscopy (XPS) (Figure 3a,b) shows the peaks related to Fe^{3+} ($\sim 711.3 \text{ eV}$), Fe^{2+} ($\sim 709.8 \text{ eV}$), and iron

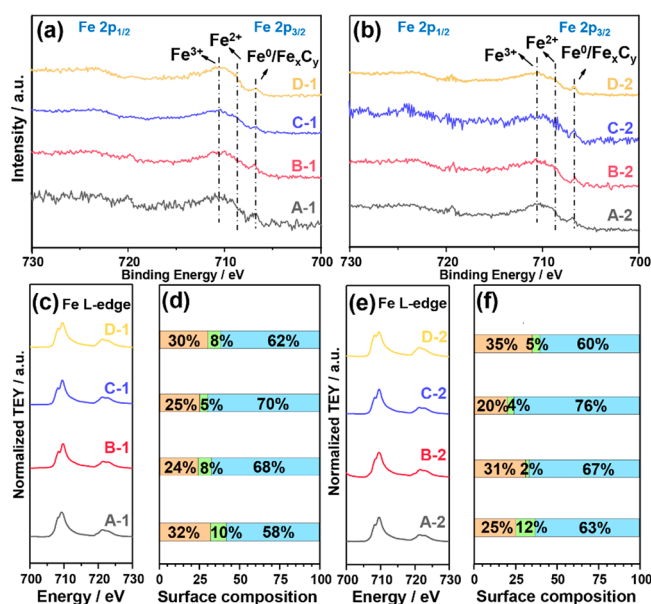


Figure 3. Surface characterization of catalysts under two conditions: (a,b) XPS spectrum of catalysts at two conditions, (c,e) Fe $\text{L}_{2,3}$ -edge X-ray absorption spectra, and (d,f) phase composition of the obtained samples determined by LCF method; in the bar graphs orange represents $\text{Fe}^{2+/3+}$, green represents Fe^{2+} , and blue represents Fe^0/FeC_x .

carbide species ($\sim 707.0 \text{ eV}$), meaning the surface of all the samples consist of iron oxide and iron carbide species, which are also confirmed by the Fe L-edge analysis. Together with previous XRD and XPS analysis, we propose the models (Figure 2d,i) for catalysts that are derived from oxidized and reduced iron precursors. Also, the bulk and surface species confirmed are of great significance to be employed as references for latter quantitative calculations.

Soft X-ray absorption spectroscopy was employed to characterize further the surface species of samples. The pre-edge of the O K -edge spectra at $\sim 532 \text{ eV}$ (Figure S2) indicates the presence of iron oxide species.^{25,26} The distinct splitting of the L_3 and L_2 edges by $\sim 1.5 \text{ eV}$ is due to the energy difference of the e_g level and t_{2g} level (Figure 3c,e).²⁷ We note that the intensity ratios of the t_{2g} feature to the e_g feature in the L_3 -edge for the samples are higher than that of the standard spectrum of the Fe_2O_3 L -edge but lower than that of the metallic iron reported in the literature, which indicates the presence of hybridized iron species.^{28–30} On the basis of the XPS and EELS results, quantitative analysis was performed using $\alpha\text{-Fe}$, FeO , and Fe_3O_4 as references for Fe^0 , pure Fe^{2+} , and mixed ($1/2$) $\text{Fe}^{2+}/\text{Fe}^{3+}$ (Figure S3), with the characteristic main L_3 -edge contribution located at 709.3 eV for Fe^{3+} , 707.9 eV for Fe^{2+} , and 706.8 eV for $\text{Fe}^0/\text{Fe}_x\text{C}_y$.³¹ A similar linear combination fitting (LCF) method was also employed on Fe-based catalysts by de Smit et al.²⁵ The relative surface compositions are shown in Figure 3d,f. It is found that the average surface iron oxidation state (OS) of samples (Figure 4) stay in a narrow

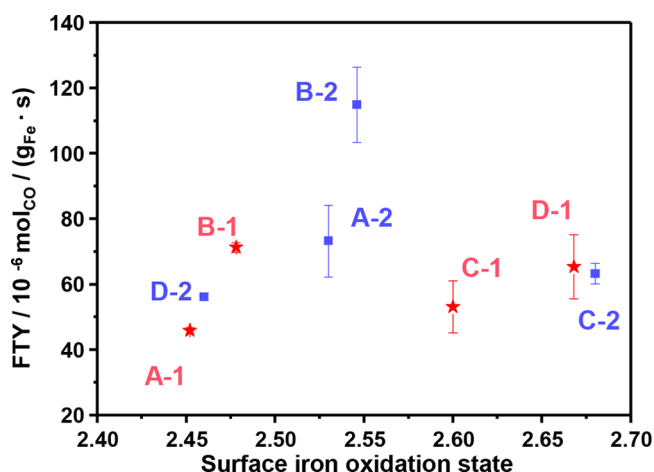


Figure 4. Catalysts FTY under different conditions as a function of surface iron OS.

range between $+2.40$ and $+2.70$. Conversely, the iron time yield (FTY, moles of CO converted to hydrocarbons per mol of Fe per second) varies greatly from 45.9 to $114.9 \times 10^{-6} \text{ mol CO g}_{\text{Fe}}^{-1} \text{ s}^{-1}$. Therefore, the surface OS cannot be used to interpret the variation of FTY reliably (Figure 4). During the FTS, the heterogeneous structure is formed under syngas atmosphere, which involves (i) CO dissociation and $\text{Fe}-\text{C}$ bond formation,^{32–34} (ii) carbon permeation and diffusion from surface to matrix,^{35,36} as well as (iii) structure reconstruction and phase transformation.²¹ Despite the controversy concerning the actual active sites, the correlation between the carbide content and activity had been widely reported, suggesting the important role of bulk properties in determining FTS performance.^{7,11,37} Raupp et al. proposed

that the bulk iron functioned as an obtainer for surface carbon, which influenced surface structure and, hence, the active site density.³⁸ In contrast, a competition model was proposed that surface “C-containing” species hydrogenation and iron carburization occur simultaneously as competitive reactions.³⁹ Bartholomew and co-workers reported that the initial catalytic activity is proportional to the amount of adsorbed atomic carbon on the surface, while the steady-state activity is related to carbon species at the surface and in the bulk of iron crystallites.⁴⁰ In recent work, Hoffmann and Wen et al. explored carbon permeation theoretically on different model surfaces and elucidated how it affects the Fe-based FTS, and carbon surface migration and subsurface permeation were found to take place simultaneously, but showing different preferences at various coverages on different crystal planes.³⁶ Although the link between surface and bulk phase of iron is still not clear, it is reasonable to postulate that both the surface and bulk properties should be taken into consideration for Fe-based FTS.

Recognizing this, we focus on analyzing the catalytic relevance of the bulk properties in these samples in the following sections. It is well acknowledged that there is a linear relationship between the average OS and the characteristic edge positions in the XANES spectra.^{41,42} Therefore, we employed several iron species as references (Figure S4) to correlate quantitatively the OS of Fe species and the characteristic edge positions. The calculated bulk OSs of two series of catalysts are summarized in Table S2. Moreover, based on previous surface and bulk characterizations analysis, Fe_3O_4 , FeC_x , and Fe_5C_2 are used as standards to calculate the relative compositions for samples from the XANES spectra (the details are provided in the Supporting Information). The phase map distribution results are displayed in Figure 5a–h. The quantitative analysis shows that most Fe species exist as carbides; only a small fraction of the Fe_3O_4 phase (<6.0%) was

detected. Moreover, the catalysts derived from various precursors are primarily identified through differences in their carbide species composition.

The catalytic testing results are summarized in Table 1. A general trend is observed: the higher the OS of iron species, the higher the FTY that can be achieved. Although the FTS has been recognized as a size-sensitive reaction, the particle sizes of iron species for all samples in our study are beyond the critical value (7 nm for an undoped catalyst).¹³ Therefore, the differences in activity and selectivity may originate from differences in their electronic properties. Notably, in Figure 6a, a linear relationship between FTY and bulk OS can be established, while the FTY versus surface OS does not reveal any clear trends (Figure 4). The OS of iron species can be considered as an indicator of the electron density of the active center, which would play an important role in the catalytic reaction. In this case, iron species with higher OS tend to be in the electronically deficient state,⁴³ possibly giving rise to an enhanced σ -coordination and a weakened π -back-donation for CO adsorption. As the contribution of π -back-donation to CO adsorption on Fe species is predominant which is confirmed via a red-shift in the Fourier Transform Infrared Spectroscopy,⁴⁴ it is reasonable that the higher OS results in a decreased strength of Fe–CO binding. On the basis of both experimental and DFT calculations,^{13,45} the weaker Fe–C bond could enhance FTS activity because of easier removal of C and O atoms from the surface. Moreover, numerous reports have revealed that χ - Fe_5C_2 , Fe_7C_3 , and Fe_2C species with higher iron OSs (+1.60, +1.71, and +2.0, respectively) are more active than carbides with lower iron OSs (i.e., Fe_4C of +1.0 and θ - Fe_3C of +1.33), which is also in line with our results.^{7,9,12,37} Therefore, although the attribution of active species is still disputable and multisite active centers may coexist, the OS of Fe species can be used as a universal descriptor to reflect the activity of CO hydrogenation.

On the other hand, as a least desired product in FTS, the formation of CH_4 via successive hydrogenation usually provides an indication of performance of the hydrogenation ability of a catalyst with respect to C–C coupling. We tried to correlate the FTY_{CH_4} to the chemical properties of Fe species, and fortunately it was found that the FTY_{CH_4} increases proportionally to the Fe–C coordination number (CN) (as displayed in Figure 6b). This result is in line with the theoretical study by Jiao's group, where a linear relationship between the effective barrier of CH_4 formation and d-band center of the surface was established.⁸ Generally, CH_4 formation can be understood as successive insertion of H atoms into the Fe–C bond. According to the principle of microscopic reversibility, the CH_4 formation thus could be easily understood by the activation of C–H, which is mainly guided by electron back-donation from Fe atoms into the antibonding σ_{CH^*} state of CH_x . In an iron carbide dominated catalyst system, as the Fe–C CN increases, the d-band center moves away from the Fermi energy, resulting in an enhanced C–H activation. Consequently, a lower effective barrier of CH_4 formation is then anticipated. Recently, the effective barrier differences between CH_4 formation and C_1 – C_1 coupling (ΔE_{eff}) on different iron phases were calculated to describe CH_4 selectivity. The ΔE_{eff} follows the order: $\text{Fe} > \text{Fe}_3\text{C} > \text{Fe}_5\text{C}_2$,^{46,47} indicating the improved CH_4 selectivity with increasing Fe–C CN, which agrees with our work as well. Theoretical calculations revealed a linear relationship between

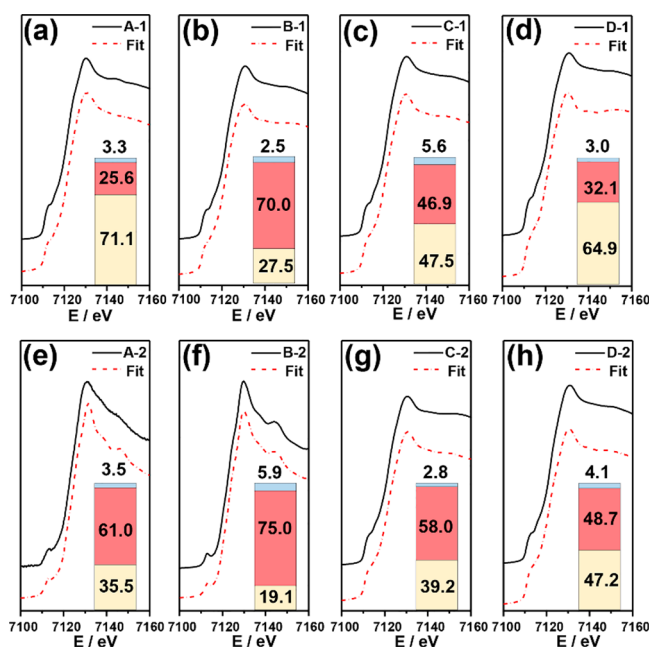


Figure 5. Fe K-edge XANES spectra and phase composition determined by LCF method of the samples obtained under condition 1 (a–d) and 2 (e–h); in the bar graphs yellow stands for Fe_3O_4 , red stands for FeC_x , and cyan stands for Fe_5C_2 .

Table 1. Catalytic Performance and Calculated Iron OS of Catalysts at Two Conditions

Sample	FTY (10^{-6} mol _{CO} g _{Fe} ⁻¹ s ⁻¹)	X_{CO} (%)	CO ₂ Sel. (%)	Hydrocarbon Sel. (%)						Average iron OS
				CH ₄	C ₂₋₄ ^a	C ₂₋₄ ⁼	C ₅ ⁺	Oxy. ^a	O/P ^b	
A-1	45.9 ± 0.9	23.9	15.2	5.0	5.1	6.3	83.6	0.0	1.2	1.21
B-1	71.3 ± 1.4	66.5	32.5	8.4	19.1	20.5	51.5	0.5	1.1	1.62
C-1	53.1 ± 8.0	32.1	35.8	9.6	24.5	11.2	54.5	0.2	0.5	1.39
D-1	65.3 ± 9.8	8.1	5.3	3.4	4.3	2.1	90.2	0.0	0.5	1.25
A-2	73.2 ± 11.0	44.1	36.2	9.7	19.4	13.5	57.4	0.0	0.7	1.82
B-2	114.9 ± 11.5	24.3	20.7	5.5	18.8	14.4	60.9	0.4	0.8	2.00
C-2	63.3 ± 3.2	49.2	36.1	9.3	16.1	15.9	58.7	0.0	1.0	1.57
D-2	56.2 ± 1.1	48.6	36.6	8.9	17.2	19.4	54.5	0.0	1.1	1.44

^aOxygenates. ^bThe molar ratio of olefin to paraffin in the C₂₋₄ hydrocarbon.

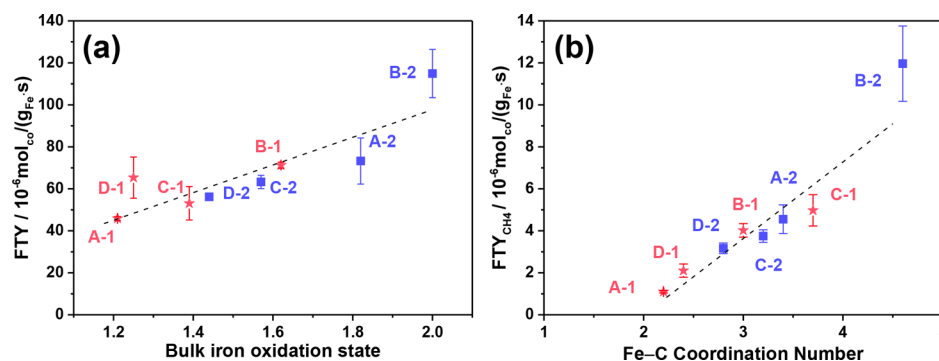


Figure 6. (a) FTY of catalysts as a function of bulk iron OS; (b) iron time yield of CH₄ (FTY_{CH₄}) as a function of Fe–C CN.

the effective barrier of CH₄ formation and the d-band center of the iron carbides species (i.e., Fe₂C (011), Fe₃C₂(010), Fe₄C (100), and Fe₃C (001), respectively),⁸ which are in accord with our results. de Jong et al. also found that increasing the electron density of iron species by introducing a promoter tended to produce less CH₄ and more lower olefins.¹⁴

Products distribution is highly dependent on the FTS CO conversion (X_{CO}), thus a fair comparison was made among catalysts under similar X_{CO} . With an X_{CO} of about 45%, A-2, C-2, and D-2 samples possess lower olefins selectivity of 13.5%, 15.9%, and 19.4%, respectively, and the relatively lower OS seems to play an important role in achieving a higher O/P molar ratio and lower olefins selectivity. This might be attributed to the weakened olefins adsorption and suppressed secondary hydrogenation on the higher electron-rich iron species, which have been confirmed by lower olefins (e.g., C₂H₄, C₃H₆) injection/pulse experiments on Fe catalysts with alkali promoters.^{48,49} These phenomena are also in good agreement with DFT calculations, which suggests that weak adsorption of olefins caused by electron donation inhibits their further hydrogenation, leading to high selectivity.⁴⁸ Above all, this phenomenon would be of great interest in our future studies.

CONCLUSION

In conclusion, eight samples produced from four pure Fe phases were obtained by changing the gaseous atmosphere and pressure. Multiple surface and bulk characterization techniques were employed to elucidate the impact of the chemical environment upon Fe species and on FTS performance. The core–shell structures are identified for all catalysts by TEM combined with EELS, but the compositions for different counterparts are strongly dependent on the properties of the iron precursors. On the basis of a precise quantification of the

different Fe related species, we found that catalytic activity and selectivity are generally governed by the bulk chemical environment of the Fe species rather than their surface properties. Despite different spatial contributions of the Fe species, the FTY is found to correlate linearly with the OS of the Fe species and the CH₄ yield is related to the CN of the Fe–C bond. Our work provides guidance for designing and modifying Fe-based FTS catalysts.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acscatal.9b01104.

Experimental section, particle size, XANES and EXAFS analysis, HRTEM (PDF)

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Notes

The authors declare no competing financial interest.

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